

Immunodeficiency, dioxins, stress, and the hormones

*There are many toxins which modify hormonal responses, activating cells and altering the immune system (including estrogens and dioxins.) When these act early in life, extremely small amounts can cause life-long changes. *When respiratory energy production is blocked in stimulated cells, the cells are likely to die. (Cortisol, estrogen, polyunsaturated oils have this effect, especially on thymus cells.) *Antibodies are involved in removing the debris of cells that have disintegrated. Intense cellular damage causes many "autoantibodies" to be produced. People with AIDS have a high incidence of "autoimmunity." *Endogenous retroviruses are activated by toxins known to be associated with immunodeficiency. Everyone has endogenous retroviruses. The antibodies which are used to diagnose "HIV" infection can, in the demonstrated absence of that virus, be produced in connection with lupus, Sjogren's syndrome, and arthritis. These autoimmune conditions are promoted by estrogen. *Estrogen activates the production of cortisol, and damages the normal feedback control, causing both cortisol and ACTH to be elevated. *Estrogen causes chronically elevated free fatty acids, and synergizes with unsaturated fats. *Estrogen inhibits thyroid function. Hypercortisolism is typically associated with hypothyroidism, and both tend to cause the loss of lean body mass. *AIDS is often compared to Addison's disease, because of hyponatremia (loss of sodium) and fatigue. Hypothyroidism causes hyponatremia and many other features seen in AIDS. *Increased levels of cortisol, estrogen, and polyunsaturated fatty acids, and decreased levels of the active thyroid hormone (T3) and (placental) progesterone have been found to occur in AIDS. *Progesterone can contribute to the inhibition of HIV replication and transmission. *Common environmental factors can produce hormonal changes leading to immunodeficiency. One hospital in southern Vietnam admitted 437 septicemia patients between mid-1993 and 1994; 23% of the adults died. In 8 months, 17,000 seals died of infections in Europe. In California, many seals die with an unusual form of metastatic cancer. Seals are highly contaminated with industrial dioxins. In Africa, aflatoxin is strongly associated with immunodeficiency. In animals, both dioxin and aflatoxin activate the expression of viruses. Endometriosis is stimulated by dioxins. Environmental estrogens affect the immune system. It has been over ten years since I wrote about "AIDS" (e.g., "Repairing the Immune System," in Cofactors in AIDS and HIV infection, edited by R.R. Watson, 1989) and the official doctrine that it is caused by the "HIV" virus still hasn't been supported by anything that resembles real science. Duesberg's arguments have never been answered (except by bureaucratic thuggery). In 1989 I pointed out that septicemia, blood stream infection, in young adults, which used to be a rare thing, and which indicates defective immunity, has been increasing in a remarkably continuous way since the late 1940s, and I reviewed the many things in our environment that are known to suppress immunity, and which have become increasingly prevalent in our environment --unsaturated vegetable oils, ferrous iron and carrageenan in our foods, lead in air, food, and water, exposure to medical, military, and industrial ionizing radiation, vaccinations, pesticides, chlorinated hydrocarbons, nitric oxide (smog and medications) and oral contraceptives and environmental estrogens, in particular. Of these factors, only radiation and lead exposure have decreased in the last several years, after several decades of rapid increase. The widespread use of diuretics in pregnancy, which began in the 1950s and contributed to an epidemic of premature births, also declined after the late 1960s. Most of these environmental factors damage the thymus gland, which regulates the immune system, and by acting on the thymus their effects tend to be additive with other immunosuppressive factors, including cancer, traumatic injury, inflammation, toxins in spoiled food (e.g., aflatoxins) and malnutrition. Cancer, AIDS, and extreme hypothyroidism have several features in common--they cause tissue loss and organ damage, with immunodeficiency and intense activation of the stress hormones, including cortisol. In cancer and AIDS, a good case has been made for the primacy of stress-induced wasting as the main cause of death. Whatever one might believe to be the cause of cancer and AIDS, it is always good for the patient to prevent tissue damage from the stress associated with the sickness. Since the stress hormones primarily destroy tissues by the activation of specific proteases, the use of protease inhibitors for treating AIDS could conceivably be affecting the stress response. However, the body's normal protection against the

cortisol-activated proteases is centered on the protective hormones, progesterone, thyroid, and the androgens. Environmental stress One of the most broadly substantiated principles in biology is that a great variety of harmful causes all lead to a few forms of biological harm--the concept of the stress reaction shows the powerful implications of the principle. Stress, no matter what the specific cause, has a particularly destructive effect on three organ systems : The nervous system, the immune system, and the reproductive system. Inflammation, lipid peroxidation, tissue atrophy, the "calcium catastrophe" (when almost anything goes wrong, calcium can transmit and amplify and extend the problem, but isn't itself the source of the problem), mitochondrial decay, and similar events help to define the stress reaction in greater detail. Hans Selye showed that the thymus shrinks very early in the stress reaction. In his understanding of the process, when adaptation was followed by the "exhaustion phase," the adrenal glands had simply become exhausted from overuse. F. Z. Meerson's work showed that cortisol, and the free fatty acids mobilized by stress, have a toxic influence on the mitochondrial energy production system. Both cortisol and the free fatty acids block the efficient use of glucose for producing energy, creating a diabetes-like condition. The exhaustion problem caused by excessive stress is generalized, not just a matter of adrenal insufficiency. Meerson's work created the basis for understanding several degenerative processes, especially the phenomenon of "excitotoxicity," in which the combination of excessive stimulation and deficient energy supply damages or kills cells. Selye believed that some hormones are antagonistic to each other. A few of the oppositions that he identified have been thoroughly researched, especially the catabolic/anabolic functions of glucocorticoids and androgens, and the shock/antishock functions of estrogen and progesterone, respectively. Puberty, because of hormonal changes, especially increased estrogen, can be seen as the first stage of a chronic stress, resembling diabetes, since elevated free fatty acids cause "insulin resistance," with slightly impaired oxidation of glucose. The thymus shrinks considerably at puberty, under the influence of the hormonal changes and the increased free fatty acids (caused mainly by estrogen). The degenerative diseases can be seen as the cumulative result of stress, in which tissue damage results from the diabetes-like impairment of energy production. The thymus, and the thymus-dependent areas of the spleen, are required for full and subtle control of immunity. In the absence of thymic control, the B cells are still able to produce antibodies, but they are more likely to produce autoantibodies. Stress produces a variety of cellular changes, including the production of the "shock proteins." These proteins can make up 20% of the cell's total protein content. In themselves, the shock proteins are immunosuppressive. They can be recognized by the immune system as antigens, and so are a factor in the appearance of "autoimmune" antibodies. The autoantibodies themselves are often blamed for the diseases they are sometimes associated with, but since they can be present (for example, following removal of the spleen) in people who have no symptoms, their function is probably to facilitate the removal of tissues which are defective for some other reason. The shock proteins could be one of the signals that activate the immune system to remove damaged tissue, and they might be involved in the removal of senescent cells, though I don't think any experiments have been done to test this idea. Besides activating the cells to produce massive amounts of the shock proteins, stress can also activate the so-called hormone receptors, such as estrogen receptors, even in the absence of the hormones. Stress also activates the endonucleases, which cut sections out of the DNA molecules, and activates mobile genetic elements, producing genetic instability. Like cortisol and estrogen, stress itself activates integrated retroviruses. The "endogenous retroviruses" make up nearly 10% of the human genome, and many of them locate themselves in regulatory sites in the chromosomes. Since stress lowers the discriminatory ability of the immune system, and stimulates the expression of retroviruses, the antibodies sometimes seen in association with immunodeficiency may be similar to the various autoantibodies that are also produced by stress. People who have autoimmune diseases such as lupus and Sjogrens syndrome (which are promoted by estrogen : Ahmed and Talal) have antibodies which sometimes react positively in the AIDS test, and searches for the HIV virus in such people have found no evidence of it. (Nelson, et al., 1994 ; Deas, et al., 1998.) Treatments for roundworms and other parasites cause antibodies to retroviruses to appear in animals that previously tested negative ; this might account for the high rates of positive tests for HIV in areas such as Africa in which treatment for filariasis is common (Kitchen and Cotter, 1988). Organisms are most sensitive to environmental damage early in life, especially prenatally. This is the period in which normal hormone exposure masculinizes the brain, for example. The term "imprinting" refers to the extreme responsiveness of the organism at this time, and it has been extended to include

long lasting influences which may result from abnormally high or low levels of natural substances, or from the presence of other, abnormal substances during the sensitive period. The effects of early "imprinting" can cause permanently altered sensitivities. In animal studies, L. C. Strong showed that prenatal influences determine the age at which puberty and reproductive senescence occur. In humans, premature birth, a powerful stressor, is associated with premature puberty. The thymus is damaged both by premature birth and by puberty. The effects of damage early in life will increase vulnerability in subsequent decades. When babies are imprinted by the mother's disturbed hormones, or by diuretics, by milk substitutes, or by industrial effluents, the worst effects are likely to be seen decades later, or even generations later. A similar long-range effect can be produced by nutritional deficiencies. Although more mature organisms are less sensitive to stress, both early imprinting, and the cumulative effects of exposure, will cause some individuals to be much more sensitive than others, and aging itself increases vulnerability. If the present epidemic of immunodeficiency is produced by environmental stress, then we should expect to see a variety of other stress-related diseases increasing at roughly the same time. When a stressor is acting through imprinting, then the harmful effects may not be seen until 20 or 30 years later, but when the stressor has acute and immediate effects, the effects should rise and fall at roughly the same time as the environmental cause. The rise of the Acquired Immunodeficiency Syndrome during the last 50 years hasn't been the only health problem that has grown rapidly during that time. The "flesh eating bacteria," causing necrotizing fasciitis and related conditions, should probably be classed along with septicemia/bacteremia as the consequence of a weakened immune system, but there are many other diseases that have followed a similar pattern, which might be caused by the same factors which are causing immunodeficiency. Thyroid diseases (mostly in women), some autoimmune diseases including primary biliary cirrhosis (mostly in women) and inflammatory bowel disease, liver cancer, diabetes (doubling in children since 1949), prostate cancer, decreased sperm counts, premature births and birth defects, minimal brain dysfunction-attention deficit-hyperactivity, cerebral palsy, premature puberty (which is associated with premature birth), congestive heart failure, osteoporosis (independently of the changing age-structure of the population), depression (most common in women, more than doubling among children in recent decades), and multiple sclerosis have increased in prevalence during this period. Some of these conditions are strongly associated with each other, for example, primary biliary cirrhosis, breast cancer, and osteoporosis. It is common knowledge, among people who study immunity, that radiation, polyunsaturated fatty acids, estrogens, and dioxins are toxic to the thymus gland, and can produce immuno-deficiency. They mimic or accelerate the thymic atrophy of aging, causing a deficient thymus-dependent immune response, usually without harming the ability of B cells to produce antibodies. There are probably many examples of damage to immune systems, besides immunodeficiency, caused by these agents. Slight damage to the immune system, such as can be produced by hypoglycemia or other energy deficit--creates an exaggerated inflammatory response, and the release of the mediators of inflammation, including histamine, serotonin, and prostaglandins, activates the stress hormone system, leading to further biological damage. Liver disease and several other "autoimmune" diseases involve abnormal immune responses, probably including thymic deficiency and an intensified inflammatory response. The fact that livers transplanted from female donors to male recipients are less successful than are livers from male donor transplanted into female recipients, is consistent with the idea that autoantibodies (which are far more common in women than in men) are a relatively harmless response to changes in the organs themselves. Are antiviral therapies working? Ivan Ilich, in *Medical Nemesis*, showed that historically, many diseases have had characteristic incidence curves, rising to a maximum, and then falling away to relative insignificance, independently of what people were doing as treatment or prevention. As susceptible people are exposed to conditions that cause a disease, they will get sick, and then either die or develop resistance. The conditions which at first caused increasing disease incidence, will eventually tend to affect only children who haven't developed resistance. If AIDS mortality rose rapidly to a peak a few years ago, and then began falling, we should ask whether this pattern fits that of other diseases discussed by Ilich. Looking for causes other than the virus, we might find a parallel in the rise and fall of some other factor. In the 1950s, new diuretics came on the market, and millions of pregnant women took them. It was predicted that there would be an epidemic of brain damage as a result, and in fact the incidence of hyperactivity, attention-deficit, and other "minimal" brain damage disorders did rise during those years. After about 15-20 years, experiences such as the Thalidomide episode caused physicians to temper their enthusiasm for the use of drugs

during pregnancy. The incidence of low birth-weight babies in the U.S. peaked around 1965, and 28 years later AIDS mortality in the US peaked. The rising curve had followed both the increase in radioactive fallout from atmospheric testing of large numbers of atomic bombs up to 1963, and the intense promotion of the new diuretics beginning in the early 1950s. The peak in AIDS mortality in 1993 came ten or twelve years after the long decline in SAT scores had stopped. (The most extreme declines in SAT scores had occurred among the brightest students, disproving the contention that the average score fell simply because more students were taking the tests.) The same prenatal damage which caused the extreme decline in SAT scores 18 years later (when the damaged babies reached that age) would have left many of the same individuals with weakened immune systems, which would fail prematurely, but at varying intervals, depending on the exposure to other factors. The use of unleaded gasoline increased into the 1990s, and there was a corresponding decrease in tissue lead content, reflecting the smaller amount of lead being put into the environment. According to some reports, medical and dental x-ray exposures were declining during this period. Yet other factors, including dioxins and unsaturated dietary fats, were probably increasing. Although the new protease inhibitors wouldn't be used until years after the AIDS mortality had begun falling, the government and drug companies are claiming that it is the drugs which are decreasing the mortality. A Synthesis Many things in our environment are increasing the incidence of certain kinds of liver disease. The liver processes things that are ingested or that enter the blood stream after being inhaled or absorbed through the skin, so in a toxic environment it is susceptible to injury. If deprived of good nutrition or adequate thyroid hormone it is especially sensitive to toxins. The body's own estrogen is a burden on the liver, causing women's livers to be on average slower than men's in processing environmental chemicals. Almost any kind of toxin causes the liver to be less efficient at excreting other substances, including hormones. In malnutrition, sickness, and in aging, there is a tendency for higher levels of estrogen to remain circulating in the blood. Natural estrogen, and environmental substances that act like estrogen, act as excitants in many types of cell, and at the same time, reduce the efficiency of energy production. Both of these properties relate to its known ability to activate the adrenal glands. A. L. Soderwall, who was my thesis adviser at the University of Oregon, found that estrogen caused hamsters' adrenal glands to enlarge, and that larger doses overstimulated the glands sufficiently to cause tissue damage. It is now known that estrogen acts directly on the adrenal cells to stimulate cortisol production, and that it also stimulates the pituitary to produce more adrenocorticotropin (ACTH), which also stimulates the adrenals; estrogen's effect is to impair the negative feedback, in which cortisol normally shuts down ACTH production. This impaired feedback is characteristic of aging. Estrogen directly causes the thymus gland to atrophy, and several of its effects, such as increased adrenal activity and elevated free fatty acids, also contribute to the shrinkage of the thymus and the inhibition of its functions. While this is happening, the B cells, which normally are under the control of the thymus cells, are not killed by estrogen, and actually seem to be stimulated by estrogen to produce certain types of antibodies. This combination of effects, weakening the thymus and stimulating antibody production, is thought to contribute to the development of autoimmune diseases. Estrogen also stimulates mast cells and similar cells to release histamine and other promoters of inflammation, and these effects are probably closer to the actual problem in the autoimmune diseases. Several of the substances formed under the influence of estrogen interfere with energy production and contribute to cellular excitation, causing tissue injury. Cortisol also stimulates antibody production while suppressing thymic immunity (Norbiato, et al., 1997). Estrogen and stress cause increased levels of free fatty acids to circulate. The polyunsaturated fatty acids are immunosuppressive, antithyroid, diabetogenic, inhibit respiration, and promote the actions of estrogen and cortisol. People suffering from AIDS have been found to have increased estrogen, with high cortisol and ACTH, and very low T3. (Unfortunately, some researchers and the editors who publish their ideas, conclude that the hormones don't cause the stress and wasting symptoms, because they call thyroid a "catabolic hormone," and because they describe the fatigue and sodium deficiency as evidence of "deficiency of cortisol." Such is the state of the research establishment.) In animal experiments, and a few human tests, the HIV and similar viruses have produced effects that could plausibly explain some of the conditions seen in AIDS, such as damage to brain cells (C. Pert, R. Sapolsky), and altered steroid secretion. But this is real science, that promises to link up with information about stress, aging, allergy, and biological adaptability. For example, Sapolsky's group (Brooke, et al., 1998) found that the nerve toxicity caused by a viral protein (called gp120) synergizes with glucocorticoid toxicity, lowering the ATP level and inhibiting mitochondrial function,

and that simply supplying the nerve with additional energy protects it from destruction. In other words, the viral peptide just increases excitotoxicity. Another group (Amirhessami-Aghili and Spector, 1991) found that the presence of the virus can decrease the production of progesterone. Since progesterone blocks (Lee, et al., 1997) the expression (and transmission) of the virus, this suggests how the overgrowth of the virus might be triggered by stress--once progesterone synthesis falls, a vicious circle could get started. Lee, et al., found that progesterone can help to prevent transmission of the virus from an infected mother to the fetus. But the most interesting study of the virus in pregnancy involved mice that were engineered to contain extremely large quantities of the HIV provirus (De, et al., 1997). At birth, they seemed normal, but within a few days their skin became diseased, and they quickly wasted away and died. The experimenters realized that something present in the mother's body had permitted normal development up to the point of birth, and then the wasting disease set in. The placental hormone, chorionic gonadotropin, is produced in large amounts during pregnancy. The experimenters gave newborn infected mice regular doses of human chorionic gonadotropin (hCG), and they developed normally. Rodents don't respond to gonadotropins or other ovarian stimulation exactly the way pigs and primates and people do. For example, prolactin and melatonin usually inhibit progesterone synthesis in people, but in rodents, they increase it. So it's necessary to see exactly what happens to the ovarian hormones when a mouse is given hCG. In 1996, another group (H. Krzanowska and M. Szoltys) had done that, and found that hCG greatly increases progesterone synthesis, but decreases estrogen. Considering the progesterone-HIV experiments together, I am reminded of a science fiction movie, in which a disease from another planet killed everyone in the lab that was studying it, except for one woman, who turned out to be pregnant. The medical version of AIDS research, though, pushes aside all of the real science, in favor of a simplistic idea that the virus kills the cells of the immune system, and uses false diagnostic methods and deadly drugs to treat something which too often doesn't exist, while denying that there are other real causes of immune deficiency and wasting-sickness, etc. Aging is characterized by loss of lean body mass, immunodeficiency, and a variety of autoimmune reactions. My perennial argument has been that decreased thyroid and progesterone, associated with increased estrogen and stress hormones, are largely responsible for those changes. The huge investment in AIDS research has found that these occur in AIDS, but, because of the medical pharmaceutical culture which has created myths about these hormones, no one is yet interpreting the hormone imbalances in ways that would reveal their responsibility for the symptoms. While the institutionalized theory claims that the HIV virus is responsible for the syndrome, the hormones are reduced to epiphenomena. REFERENCES

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